

Databricks Data Engineer Associate Exam - Practice Questions

Section 1: Databricks Lakehouse Platform (15 questions)

Question 1

What is the primary advantage of the Databricks Lakehouse architecture?

- A) It separates storage and compute completely
- B) It combines the benefits of data lakes and data warehouses
- C) It only supports structured data
- D) It eliminates the need for ETL processes

Answer: B

Question 2

Which component is responsible for metadata management in the Databricks Lakehouse?

- A) Delta Lake
- B) Unity Catalog
- C) MLflow
- D) Apache Spark

Answer: B

Question 3

What is the default file format used by Delta Lake?

- A) Avro
- B) JSON

- C) Parquet
- D) ORC

Answer: C

Question 4

Which of the following is NOT a benefit of the Databricks Lakehouse platform?

- A) ACID transactions
- B) Schema enforcement
- C) Real-time streaming only
- D) Time travel capabilities

Answer: C

Question 5

What does Unity Catalog provide in the Databricks ecosystem?

- A) Only data governance
- B) Only access control
- C) Unified governance, discovery, and access control
- D) Only metadata management

Answer: C

Question 6

Which cluster type is most suitable for production ETL workloads?

- A) All-purpose cluster
- B) Job cluster
- C) Interactive cluster
- D) Development cluster

Answer: B

Question 7

What is the recommended approach for managing secrets in Databricks?

- A) Store them in notebooks
- B) Use Databricks Secrets
- C) Hard-code them in applications
- D) Store them in plain text files

Answer: B

Question 8

Which feature allows you to query previous versions of a Delta table?

- A) Schema evolution
- B) Time travel
- C) Optimize
- D) Vacuum

Answer: B

Question 9

What is the purpose of the OPTIMIZE command in Delta Lake?

- A) To delete old files
- B) To compact small files and improve query performance
- C) To update table schema
- D) To create checkpoints

Answer: B

Question 10

Which workspace object is used to store and organize code in Databricks?

- A) Jobs
- B) Clusters
- C) Notebooks
- D) Dashboards

Answer: C

Question 11

What is the maximum number of nodes recommended for an interactive cluster?

- A) 10
- B) 20
- C) 30
- D) 50

Answer: A

Question 12

Which authentication method is recommended for production applications?

- A) Personal Access Tokens
- B) Username/Password
- C) Service Principal
- D) Anonymous access

Answer: C

Question 13

What happens to job clusters after job completion?

- A) They remain running
- B) They are automatically terminated

- C) They enter standby mode
- D) They are converted to all-purpose clusters

Answer: B

Question 14

Which feature helps reduce costs by automatically scaling clusters?

- A) Auto Loader
- B) Autoscaling
- C) Auto Optimize
- D) Auto Vacuum

Answer: B

Question 15

What is the recommended way to share code across multiple notebooks?

- A) Copy and paste
- B) Use %run magic command
- C) Store in external files
- D) Use email attachments

Answer: B

Section 2: ELT with Spark SQL and Python (25 questions)

Question 16

Which SQL command is used to create a new Delta table?

- A) CREATE TABLE USING PARQUET
- B) CREATE TABLE USING DELTA

- C) CREATE DELTA TABLE
- D) CREATE TABLE USING DATABRICKS

Answer: B

Question 17

What is the correct syntax to read a CSV file using PySpark?

- A) spark.read.csv("path")
- B) spark.read.format("csv").load("path")
- C) Both A and B
- D) spark.csv.read("path")

Answer: C

Question 18

Which transformation is used to add a new column to a DataFrame?

- A) select()
- B) withColumn()
- C) addColumn()
- D) newColumn()

Answer: B

Question 19

What does the collect() action do?

- A) Returns the first row
- B) Returns all rows as an array to the driver
- C) Counts the number of rows
- D) Writes data to storage

Answer: B

Question 20

Which method is used to write data in append mode?

- A) `.mode("append").save()`
- B) `.append().save()`
- C) `.writeMode("append")`
- D) `.save("append")`

Answer: A

Question 21

What is the purpose of the MERGE statement in Delta Lake?

- A) To join two tables
- B) To perform upsert operations
- C) To merge schemas
- D) To combine partitions

Answer: B

Question 22

Which function is used to handle null values in Spark SQL?

- A) `IFNULL()`
- B) `COALESCE()`
- C) `ISNULL()`
- D) Both A and B

Answer: D

Question 23

What is the difference between `cache()` and `persist()`?

- A) No difference
- B) `cache()` uses `MEMORY_ONLY`, `persist()` allows storage level choice
- C) `cache()` is faster
- D) `persist()` is deprecated

Answer: B

Question 24

Which window function is used to assign row numbers?

- A) `ROW_NUMBER()`
- B) `RANK()`
- C) `DENSE_RANK()`
- D) `NTILE()`

Answer: A

Question 25

What does the `explode()` function do?

- A) Splits a string
- B) Creates multiple rows from array/map columns
- C) Removes duplicates
- D) Aggregates data

Answer: B

Question 26

Which join type returns all rows from both tables?

- A) `INNER JOIN`
- B) `LEFT JOIN`

- C) RIGHT JOIN
- D) FULL OUTER JOIN

Answer: D

Question 27

What is the correct way to filter data in PySpark?

- A) df.filter()
- B) df.where()
- C) Both A and B
- D) df.select()

Answer: C

Question 28

Which method is used to remove duplicate rows?

- A) unique()
- B) distinct()
- C) removeDuplicates()
- D) deduplicate()

Answer: B

Question 29

What does the pivot() operation do?

- A) Rotates data from rows to columns
- B) Rotates data from columns to rows
- C) Sorts data
- D) Filters data

Answer: A

Question 30

Which function is used to convert string to timestamp?

- A) to_timestamp()
- B) cast()
- C) Both A and B
- D) timestamp()

Answer: C

Question 31

What is the purpose of broadcast joins?

- A) To improve performance for small tables
- B) To reduce shuffle operations
- C) Both A and B
- D) To increase parallelism

Answer: C

Question 32

Which aggregation function calculates the standard deviation?

- A) stddev()
- B) std()
- C) stdev()
- D) standard_deviation()

Answer: A

Question 33

What does the repartition() method do?

- A) Increases the number of partitions
- B) Decreases the number of partitions
- C) Can increase or decrease partitions
- D) Only sorts partitions

Answer: C

Question 34

Which method is used to write data to multiple output formats?

- A) multiWrite()
- B) Multiple write statements
- C) write().format().save()
- D) writeMultiple()

Answer: B

Question 35

What is the difference between count() and show()?

- A) count() returns number of rows, show() displays data
- B) No difference
- C) count() is an action, show() is a transformation
- D) show() is faster

Answer: A

Question 36

Which SQL clause is used to limit the number of rows returned?

- A) TOP
- B) LIMIT

C) HEAD

D) FIRST

Answer: B

Question 37

What does the when() function do in PySpark?

A) Filters data

B) Creates conditional expressions

C) Joins tables

D) Sorts data

Answer: B

Question 38

Which method is used to union two DataFrames?

A) union()

B) unionAll()

C) combine()

D) Both A and B

Answer: D

Question 39

What is the purpose of checkpointing?

A) To save intermediate results

B) To truncate lineage

C) To improve performance

D) All of the above

Answer: D

Question 40

Which function is used to generate unique identifiers?

- A) monotonically_increasing_id()
- B) uuid()
- C) unique_id()
- D) generate_id()

Answer: A

Section 3: Incremental Data Processing (20 questions)

Question 41

What is Auto Loader primarily used for?

- A) Automatic data loading from cloud storage
- B) Incremental file processing
- C) Schema inference and evolution
- D) All of the above

Answer: D

Question 42

Which format does Auto Loader use to track processed files?

- A) Delta format
- B) Checkpoint format
- C) JSON format
- D) Parquet format

Answer: B

Question 43

What is the recommended approach for handling late-arriving data?

- A) Ignore it
- B) Use watermarks
- C) Reprocess all data
- D) Store in separate table

Answer: B

Question 44

Which trigger type processes data as soon as it arrives?

- A) Once trigger
- B) Continuous trigger
- C) Processing time trigger
- D) Available now trigger

Answer: B

Question 45

What does the maxFilesPerTrigger option control?

- A) Maximum file size
- B) Number of files processed per batch
- C) Maximum processing time
- D) Number of partitions

Answer: B

Question 46

Which method is used to start a streaming query?

- A) start()
- B) begin()
- C) execute()
- D) run()

Answer: A

Question 47

What is the purpose of the MERGE statement in streaming?

- A) To handle updates and inserts
- B) To join streams
- C) To merge schemas
- D) To combine partitions

Answer: A

Question 48

Which option enables exactly-once processing guarantee?

- A) checkpointLocation
- B) outputMode
- C) trigger
- D) watermark

Answer: A

Question 49

What does the WHEN NOT MATCHED clause do in MERGE?

- A) Handles updates
- B) Handles inserts

- C) Handles deletes
- D) Handles errors

Answer: B

Question 50

Which cloud notification service does Auto Loader support?

- A) AWS SQS
- B) Azure Event Grid
- C) Google Cloud Pub/Sub
- D) All of the above

Answer: D

Question 51

What is the default behavior when schema changes occur in Auto Loader?

- A) Fail the stream
- B) Ignore new columns
- C) Add new columns as strings
- D) Stop processing

Answer: C

Question 52

Which output mode appends new records only?

- A) complete
- B) update
- C) append
- D) upsert

Answer: C

Question 53

What is the purpose of watermarks in streaming?

- A) To handle late data
- B) To define event time
- C) To control memory usage
- D) All of the above

Answer: D

Question 54

Which option specifies the location for checkpoint files?

- A) checkpointPath
- B) checkpointLocation
- C) checkpointDir
- D) checkpoint

Answer: B

Question 55

What does the cloudFiles.format option specify?

- A) Output format
- B) Input file format
- C) Checkpoint format
- D) Schema format

Answer: B

Question 56

Which method is used to stop a streaming query?

- A) stop()
- B) terminate()
- C) end()
- D) kill()

Answer: A

Question 57

What happens if you don't specify a watermark for late data?

- A) Data is processed indefinitely
- B) Late data is ignored
- C) Stream fails
- D) Data is processed with default watermark

Answer: A

Question 58

Which trigger processes all available data once?

- A) Once trigger
- B) Continuous trigger
- C) Processing time trigger
- D) Available now trigger

Answer: A

Question 59

What is the recommended pattern for handling CDC data?

- A) Use MERGE statement
- B) Use append-only pattern

- C) Use complete refresh
- D) Use delete and insert

Answer: A

Question 60

Which option controls schema evolution in Auto Loader?

- A) cloudFiles.schemaEvolution
- B) cloudFiles.schemaEvolutionMode
- C) Both A and B
- D) schemaEvolution

Answer: C

Section 4: Production Pipelines (25 questions)

Question 61

What is Delta Live Tables primarily used for?

- A) Real-time analytics
- B) Declarative ETL pipelines
- C) Machine learning
- D) Data visualization

Answer: B

Question 62

Which decorator is used to define a table in DLT?

- A) @table
- B) @dlt.table

- C) @delta.table
- D) @pipeline.table

Answer: B

Question 63

What does the LIVE keyword represent in DLT?

- A) Real-time data
- B) References to other DLT datasets
- C) Live streaming
- D) Production data

Answer: B

Question 64

Which cluster mode is recommended for DLT pipelines?

- A) Standard
- B) High Concurrency
- C) Single User
- D) DLT automatically manages clusters

Answer: D

Question 65

What is the purpose of expectations in DLT?

- A) Performance optimization
- B) Data quality constraints
- C) Schema validation
- D) Access control

Answer: B

Question 66

Which expectation action drops bad records?

- A) @expect
- B) @expect_or_drop
- C) @expect_or_fail
- D) @expect_all

Answer: B

Question 67

What does the STREAMING LIVE TABLE create?

- A) Batch processing table
- B) Real-time processing table
- C) Static table
- D) Temporary table

Answer: B

Question 68

How do you schedule a DLT pipeline?

- A) Using cron expressions
- B) Using triggers in pipeline settings
- C) Using Databricks Jobs
- D) Automatic scheduling only

Answer: C

Question 69

What is the difference between LIVE TABLE and STREAMING LIVE TABLE?

- A) No difference
- B) LIVE TABLE is for batch, STREAMING LIVE TABLE is for incremental
- C) LIVE TABLE is deprecated
- D) STREAMING LIVE TABLE is faster

Answer: B

Question 70

Which mode processes all source data every time?

- A) Triggered
- B) Continuous
- C) Production
- D) Development

Answer: D

Question 71

What does the APPLY CHANGES INTO statement do?

- A) Applies schema changes
- B) Implements CDC pattern
- C) Applies data transformations
- D) Applies security policies

Answer: B

Question 72

Which parameter specifies the keys for CDC operations?

- A) primary_keys
- B) keys

- C) sequence_by
- D) merge_keys

Answer: B

Question 73

What is the purpose of the sequence_by parameter in APPLY CHANGES?

- A) To order records
- B) To identify the sequence column for CDC
- C) To sort output
- D) To partition data

Answer: B

Question 74

Which view shows data quality metrics in DLT?

- A) system.dlt.pipeline_events
- B) system.quality_metrics
- C) dlt.data_quality
- D) information_schema.quality

Answer: A

Question 75

What does EXCEPT SELECT statement do in CDC?

- A) Excludes columns
- B) Handles delete operations
- C) Filters records
- D) Excludes tables

Answer: B

Question 76

How do you reference a table from another pipeline?

- A) Use fully qualified name
- B) Import the pipeline
- C) Use LIVE keyword
- D) Not possible

Answer: A

Question 77

What is the recommended way to handle sensitive data in DLT?

- A) Encrypt columns
- B) Use expectations to mask data
- C) Store separately
- D) Use row-level security

Answer: B

Question 78

Which storage location contains DLT pipeline metadata?

- A) Tables location
- B) Checkpoint location
- C) System tables location
- D) All of the above

Answer: D

Question 79

What does the TBLPROPERTIES clause specify in DLT?

- A) Table properties
- B) Pipeline properties
- C) Data properties
- D) Schema properties

Answer: A

Question 80

How do you handle schema evolution in DLT?

- A) Manual updates only
- B) Automatic schema evolution
- C) Use ALTER TABLE statements
- D) Recreate tables

Answer: B

Question 81

What is the purpose of the target database in DLT?

- A) Source database
- B) Destination for created tables
- C) Staging database
- D) Temporary database

Answer: B

Question 82

Which compute mode provides better cost optimization?

- A) Continuous
- B) Triggered

- C) Enhanced Autoscaling
- D) Standard

Answer: B

Question 83

What does the CONSTRAINT keyword do in expectations?

- A) Adds foreign key constraints
- B) Creates data quality rules
- C) Enforces schema
- D) Controls access

Answer: B

Question 84

How do you monitor DLT pipeline performance?

- A) System tables only
- B) Pipeline events only
- C) Both system tables and pipeline events
- D) External monitoring tools only

Answer: C

Question 85

What is the benefit of using LIVE views?

- A) Better performance
- B) Automatic dependency management
- C) Simplified syntax
- D) All of the above

Answer: D

Section 5: Data Governance (15 questions)

Question 86

What is Unity Catalog's primary purpose?

- A) Data processing
- B) Unified data governance
- C) Machine learning
- D) Visualization

Answer: B

Question 87

Which level of the Unity Catalog hierarchy contains tables?

- A) Catalog
- B) Schema
- C) Table
- D) Column

Answer: B

Question 88

What does the three-level namespace in Unity Catalog consist of?

- A) workspace.database.table
- B) catalog.schema.table
- C) database.schema.table
- D) metastore.catalog.table

Answer: B

Question 89

Which privilege allows reading data from a table?

- A) SELECT
- B) READ
- C) USE
- D) ACCESS

Answer: A

Question 90

What is a metastore in Unity Catalog?

- A) Data storage location
- B) Top-level container for metadata
- C) Type of catalog
- D) Security boundary

Answer: B

Question 91

Which command shows current user's privileges?

- A) SHOW GRANTS
- B) DESCRIBE PRIVILEGES
- C) SHOW PRIVILEGES
- D) LIST GRANTS

Answer: A

Question 92

What does row-level security provide?

- A) Column-level access
- B) Table-level access
- C) Row-level access control
- D) Schema-level access

Answer: C

Question 93

Which object type can be used to simplify privilege management?

- A) Roles
- B) Groups
- C) Service principals
- D) All of the above

Answer: D

Question 94

What is the purpose of data lineage in Unity Catalog?

- A) Performance monitoring
- B) Track data dependencies
- C) Security auditing
- D) Storage optimization

Answer: B

Question 95

Which feature provides automatic data discovery?

- A) Auto Catalog
- B) Data Discovery

- C) Automatic cataloging
- D) Smart discovery

Answer: C

Question 96

What does the USE CATALOG command do?

- A) Creates a catalog
- B) Sets the current catalog
- C) Describes a catalog
- D) Drops a catalog

Answer: B

Question 97

Which type of table is managed by Unity Catalog?

- A) External tables only
- B) Managed tables only
- C) Both managed and external tables
- D) Delta tables only

Answer: C

Question 98

What is the purpose of information_schema?

- A) Store user data
- B) Store system metadata
- C) Store application logs
- D) Store configurations

Answer: B

Question 99

Which command is used to create a new catalog?

- A) CREATE CATALOG
- B) ADD CATALOG
- C) NEW CATALOG
- D) MAKE CATALOG

Answer: A

Question 100

What does dynamic view provide in Unity Catalog?

- A) Real-time data updates
- B) Conditional data access based on user
- C) Automatic refreshing
- D) Performance optimization

Answer: B

Bonus Questions (5 additional questions)

Question 101

Which format provides ACID transactions in Databricks?

- A) Parquet
- B) Delta Lake
- C) JSON
- D) Avro

Answer: B

Question 102

What is the recommended way to handle small files problem?

- A) Use OPTIMIZE command
- B) Increase file size
- C) Use fewer partitions
- D) Use VACUUM command

Answer: A

Question 103

Which function helps identify duplicates in a dataset?

- A) duplicate()
- B) row_number() OVER (PARTITION BY ... ORDER BY ...)
- C) distinct()
- D) unique()

Answer: B

Question 104

What does Z-ordering optimize?

- A) Storage cost
- B) Query performance for specific columns
- C) Compression ratio
- D) Schema evolution

Answer: B

Question 105

Which command removes old files from Delta table?

- A) OPTIMIZE
- B) VACUUM
- C) CLEAN
- D) PURGE

Answer: B

Answer Key Summary

Section 1 (Lakehouse Platform): 1-B, 2-B, 3-C, 4-C, 5-C, 6-B, 7-B, 8-B, 9-B, 10-C, 11-A, 12-C, 13-B, 14-B, 15-B

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Bonus Questions: 101-B, 102-A, 103-B, 104-B, 105-B

Study Tips

1. **Focus on Delta Lake features:** ACID transactions, time travel, OPTIMIZE, VACUUM
2. **Master Spark SQL and PySpark:** DataFrames, transformations, actions, joins
3. **Understand streaming concepts:** Auto Loader, watermarks, triggers, checkpoints

4. **Practice DLT syntax:** @dlt.table, expectations, APPLY CHANGES INTO

5. **Learn Unity Catalog:** Three-level namespace, privileges, governance features

Good luck with your exam!